

Imagine you're a manufacturer trying to minimize the cost of producing a certain item.

The cost might depend on two variables: the amount of raw material used (x) and the number of labor hours (y).

Let's say your cost function is given by:

$$C(x, y) = x^2 + 2y^2 - 4x - 6y + xy + 100$$

Get combination of x and y that generate minimal cost.